

METALS AND NON-METALS

Chapter 6

INTRODUCTION

- Pure substance is made up of the same kind of particles e.g gold
- A mixture is a combination of two or more substances that do not react chemically and can be separated physically
- Compound is a substance made up of two or more substances that are chemically combined

Periodic table of elements

- Elements are arranged in order of increasing **atomic number**
- Vertical columns of elements with similar properties are called **group**
- Horizontal rows are called **periods**

Periodic table

Periodic Table of the Elements																			
1 1IA 1A																	18 VIIIA 8A		
1 H Hydrogen 1.00794	2 IIA 2A											13 IIIA 3A	14 IVA 4A	15 VA 5A	16 VIA 6A	17 VIIA 7A	2 He Helium 4.00260		
3 Li Lithium 6.941	4 Be Beryllium 9.01218											5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.00644	8 O Oxygen 15.9994	9 F Fluorine 18.998403	10 Ne Neon 20.1797		
11 Na Sodium 22.989769	12 Mg Magnesium 24.304	3 IIIB 3B	4 IVB 4B	5 VB 5B	6 VIB 6B	7 VIIB 7B	8 VIII 8		9 VIII 8	10 VIII 8	11 IB 1B	12 IIB 2B	13 Al Aluminum 26.981539	14 Si Silicon 28.0859	15 P Phosphorus 30.973762	16 S Sulfur 32.06	17 Cl Chlorine 35.4527	18 Ar Argon 36.96	
19 K Potassium 39.0983	20 Ca Calcium 40.078	21 Sc Scandium 44.95591	22 Ti Titanium 47.86	23 V Vanadium 50.9415	24 Cr Chromium 51.9961	25 Mn Manganese 54.938	26 Fe Iron 55.847	27 Co Cobalt 58.9332	28 Ni Nickel 58.6934	29 Cu Copper 63.546	30 Zn Zinc 65.39	31 Ga Gallium 69.723	32 Ge Germanium 72.64	33 As Arsenic 74.9216	34 Se Selenium 78.96	35 Br Bromine 79.904	36 Kr Krypton 83.80		
37 Rb Rubidium 85.4678	38 Sr Strontium 87.62	39 Y Yttrium 88.90585	40 Zr Zirconium 91.224	41 Nb Niobium 92.90638	42 Mo Molybdenum 95.94	43 Tc Technetium 98.9062	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.9055	46 Pd Palladium 106.42	47 Ag Silver 107.8662	48 Cd Cadmium 112.411	49 In Indium 114.818	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.6	53 I Iodine 126.90447	54 Xe Xenon 131.29		
55 Cs Cesium 132.90545	56 Ba Barium 137.327	57-71		72 Hf Hafnium 178.49	73 Ta Tantalum 180.9479	74 W Tungsten 183.85	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.9665	80 Hg Mercury 200.59	81 Tl Thallium 204.3833	82 Pb Lead 207.2	83 Bi Bismuth 208.98037	84 Po Polonium [209]	85 At Astatine [210]	86 Rn Radon [222]	
87 Fr Francium [223]	88 Ra Radium [226]	89-103		104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [277]	109 Mt Meitnerium [268]	110 Ds Darmstadtium [271]	111 Rg Roentgenium [272]	112 Cn Copernicium [285]	113 Uut Ununtrium [284]	114 Fl Flerovium [289]	115 Uup Ununpentium [288]	116 Lv Livermorium [293]	117 Uus Ununseptium [294]	118 Uuo Ununoctium [294]	
Lanthanide Series																			
57 La Lanthanum 138.9055 58 Ce Cerium 140.115 59 Pr Praseodymium 140.90768 60 Nd Neodymium 144.24 61 Pm Promethium 144.9127 62 Sm Samarium 150.36 63 Eu Europium 151.9658 64 Gd Gadolinium 157.25 65 Tb Terbium 158.92534 66 Dy Dysprosium 162.50 67 Ho Holmium 164.93032 68 Er Erbium 167.26 69 Tm Thulium 168.93421 70 Yb Ytterbium 173.04 71 Lu Lutetium 174.967																			
Actinide Series																			
89 Ac Actinium 227.0337 90 Th Thorium 232.0377 91 Pa Protactinium 231.03688 92 U Uranium 238.02891 93 Np Neptunium 237.04817 94 Pu Plutonium 244.0642 95 Am Americium 243.0613 96 Cm Curium 247.0754 97 Bk Berkelium 247.06715 98 Cf Californium 251.07888 99 Es Einsteinium 252.083 100 Fm Fermium 257.103 101 Md Mendelevium 258.10 102 No Nobelium 259.103 103 Lr Lawrencium 262.109																			
Alkal Metals		Alkaline Earths		Transition Metals		Basic Metals		Semi-Metals		Nonmetals		Halogens		Noble Gases		Lanthanides		Actinides	

Metals and non- metals

Properties of metals

- **Hardness**
- Conduct **Electrical** and have good **thermal conductivity**
- Have high **density**
- **Malleability** and **elasticity** -Are malleable i.e can be hammered and rolled into sheet without breaking
- Elasticity – some are able to return to their old or original shape after stretching e.g spring
- Have high **tensile strength**- i.e do not break easily under normal forces and therefore they can be stretched into wires (**ductility**)

Difference between metals and non-metals

Metals	Non-metals
Are solids at room temperature	Are liquid or gases at room temperature
Are hard and strong	Are soft and brittle
Have lustre- shiny appearance	Have dull surfaces
Have high melting point and boiling points	Have low melting and boiling points
Produce a ringing sound when struck they are sonorous	Produces no sound when struck
Have high density	Have low density
Have high tensile strength	Have low tensile strength
Are malleable	Are non-malleable
Are ductile	Are non-ductile
Are good conductors of heat and electricity	Are poor conductors of heat and electricity

Uses of metals

- Jewellery .eg gold and silver
- Construction e.g iron
- Making doors and windows e.g aluminium is used in making door frames and windows
- Coins and medals e.g silver and gold
- Thermometers e.g mercury
- Galvanizing e.g ti and zinc

Uses of non-metals

- Sulphur for making sulphuric acids
- Chlorine- for water purification
- Inert gases- e.g helium is used in weather balloons
- Neon- advertising signs
- Argon- used in electric bulbs